

Technology Stack

Date	26 June 2026
Team ID	SWTID-2026-8613
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

Define Technology Stack:

The Personalized Networking Assistant is developed using technologies that make the system simple, fast, and reliable. Streamlit provides an easy-to-use interface, FastAPI handles the backend operations, OpenAI GPT generates personalized conversation starters, and SQLite stores user information securely. Together, these technologies help the application deliver a smooth and efficient networking experience.

Technology Stack Details

S.NO	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1.	Frontend / Client-Side	Streamlit (Python)	Used to build a simple and interactive interface where users enter event details, view event themes, and receive AI-generated personalized conversation starters.
2.	Backend / Server-Side	Python, FastAPI	Manages API requests, validates user input, coordinates AI modules, performs fact-checking, stores conversation history, and returns personalized networking suggestions efficiently.

3.	Database / Data Storage	SQLite (test_pns.db)	Stores user profiles, event details, generated conversation starters, and conversation history using a lightweight relational database suitable for the current project.
4.	Cloud / Hosting / Deployment	Streamlit (Frontend), FastAPI Server	Used for deploying the frontend and backend, making the application easily accessible while supporting future scalability and maintenance.
5.	Version Control & CI/CD	Git, GitHub	Used for version control, source code management, project collaboration, and maintaining different versions of the Personalized Networking Assistant.
6.	Third-Party APIs / Other Tools	OpenAI GPT API, Pydantic	OpenAI GPT generates personalized conversation starters based on event details, while Pydantic validates request and response data to ensure reliable system performance.